

RESEARCH, EXPERIENCES, AND ACTIVITIES

Website: kaishaikh.github.io

TEACHING EXPERIENCE

Course Instructor (UofT St George campus):

Summer 2026: MAT334: Complex Variables

Teaching Assistant (UofT St George campus):

Spring 2026: MAT267: Advanced Ordinary Differential Equations

MAT401: Polynomial Equations and Fields

Fall 2025: MAT334: Complex Variables

Summer 2025: MAT223: Linear Algebra I

Spring 2025: MAT224: Linear Algebra II

MAT267: Advanced Ordinary Differential Equations

Fall 2024: MAT224: Linear Algebra II

MAT327: Introduction to Topology

Summer 2024: MAT235: Multivariable Calculus

Spring 2024: MAT224: Linear Algebra II

Fall 2023: MAT224: Linear Algebra II

Summer 2023: MAT301: Groups and Symmetries

Spring 2023: MAT136: Calculus II

Fall 2022: MAT135: Calculus I

CONFERENCES + WORKSHOPS

Mar. 2026: FRG Workshop on Higher Categories and Geometry (@NITMB)

Feb. 2026: Midwest Topology Seminar (@Case Western Reserve)

Oct. 2025: Equivariant, Motivic, and Physical Topology in the Midwest (@UMN)

Oct. 2025: Ontario Topology Seminar (@Western)

Sept. 2025: European Autumn School in Topology (EAST)

Jul-Aug 2025: Summer Research Institute in Algebraic Geometry

Apr. 2025: Graduate Student Topology and Geometry Conference (GSTGC @IU)

Dec. 2024: Life After the Telescope Conjecture (SLMath)

Aug. 2024: Talbot Workshop: Topological Cyclic Homology of Ring Spectra
Mentored by Jeremy Hahn and Allen Yuan

July 2024: PCMI Graduate Summer School: Motivic Homotopy Theory

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June 2024: Ontario Graduate Mathematics Conference (@Waterloo)
May 2024: Atlantic TQFT Spring School (@Memorial)
Oct. 2023: FRG Workshop on Brauer Groups and Derived Categories
Organized by Ben Antieau

TALKS

Sept. 2025: EAST: Abelian Categories
Apr. 2025: GSTGC: Geometric Crystallinity
Mar. 2025: Bird's Eye Conference (@UofT): Motivic Filtrations
June 2024: Ontario Graduate Mathematics Conference: What is Algebraic K-theory?
Mar. 2024: Bird's Eye Conference (@UofT): An introduction to p-adic geometry

MENTORSHIP

Directed Reading Program (University of Toronto):

Spring 2025: Introduction to p-adic Hodge Theory (following notes by Serin Hong)

Spring 2025: Introduction to Hodge Theory (following Voisin's text)

Math Mentorship Program (University of Toronto):

Spring 2026: 3 Mentees: QFT

Spring 2024: 1 Mentee: Complex Analysis

Spring 2023: 1 Mentee: Category theory and linear algebra

OTHER RESEARCH EXPERIENCES

2021-2022 Artin-Winters proof of the semi-stable reduction theorem revisited
Masters thesis (expository); completed during masters at Oxford
Supervised by Damian Rössler
University of Oxford OMMS

Summer 2020 Numerical Covers of the Modular Surface REU Participant
Supervised by Alex Kontorovich
Rutgers University DIMACS
